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ADVANCED MANAGEMENT PROGRAMME IN BUSINESS ANALYTICS

CAPSTONE PROJECT REPORT
AI-Driven Intelligent IT Operations Platform

Ticket Intelligence Engine · SLA Breach Governance · Predictive Maintenance · Procurement Optimisation

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Abstract

Ops Intel is an AI-driven IT operations management platform developed as an ISB Capstone Project for Innodatatics. The platform integrates four machine-learning modules addressing the core pain points of enterprise IT service management: a Ticket Intelligence Engine that classifies, retrieves, and routes IT service tickets using NLP and cosine-similarity search; an SLA Breach Governance model that predicts breach probability using CatBoost and triggers automated four-tier escalation; a Predictive Maintenance system using a Random Survival Forest to forecast hardware failure timing; and a Procurement Optimisation algorithm that computes demand-driven reorder quantities from survival model predictions.

The platform is trained on 22,491 service tickets, 45,109 hardware telemetry records, and 447 asset records spanning 2023–2025. Key results from executed notebooks: Ticket Category classification achieves ROC-AUC 0.951 (Logistic Regression); Issue Type achieves F1-Weighted 0.728 (Logistic Regression, 71.8% accuracy; XGBoost near-tied at 72.9%); Resolution time regression achieves $R^2 = 0.837$ (Random Forest Regressor, RMSE 513 min); SLA CatBoost AUC = 0.5058; at recall-optimised threshold (0.4922), recall = 94.7%; and the Random Survival Forest achieves C-Index = 98.90%. The integrated OpsIntel dashboard — built in React and FastAPI — provides role-based real-time operational intelligence.

Keywords

IT Operations Management, TF-IDF, Cosine Similarity, SLA Breach Prediction, CatBoost, Recall, AHP Engineer Routing, Survival Analysis, Random Survival Forest, Concordance Index, Procurement Optimisation, React, FastAPI

Executive Summary

Enterprise IT operations teams managing hundreds of tickets daily across fragmented systems face four interconnected challenges: reactive incident management, backward-looking SLA monitoring, unpredictable hardware failures, and gut-feel procurement. Ops Intel, developed by the ISB AMPBA 2025W team for Innodatatics, replaces this fragmented approach with an integrated AI platform following a seven-stage intelligence pipeline (SENSE → PROCESS → PREDICT → COORDINATE → DECIDE → ACT → LEARN). The platform is trained on three years of historical operational data from 22,491 tickets, 45,109 telemetry readings, and 447 hardware assets spanning 2023–2025.

Ticket Intelligence Engine

The NLP classification pipeline achieves Ticket Category ROC-AUC of 0.951 (Logistic Regression, 84.2% accuracy) and Issue Type F1-Weighted of 0.728 (Logistic Regression; XGBoost near-tied at 72.9% accuracy). Resolution time is predicted with $R^2 = 0.837$ (RMSE 513 min). A TF-IDF cosine-similarity retrieval engine indexes 15,363 tickets with real descriptions, surfacing the top 5 most similar historical cases. An AHP-based engineer assignment system scores 20 engineers across 5 criteria with tier-adaptive weights (low/medium/high SLA risk), with all Consistency Ratios < 0.10 confirming Saaty (1977) methodological validity. The most impactful near-term improvement requires no model retraining: enforcing mandatory “*ticket_detailed_description*” at creation will immediately close the 31.69% data gap (7,128 near-empty rows) that forces the engine to fall back from semantic NLP retrieval to structured-field matching, directly improving routing precision and engineer assignment quality.

SLA Breach Governance

Five models were evaluated for SLA breach prediction. CatBoost achieved the highest ROC-AUC of 0.5058, with recall of 43.1% at balanced threshold; at the recall-optimised threshold (0.4922), CatBoost recall reaches 94.7%. As documented in the handover, AUC near 0.50 is mathematically correct on synthetic data with uniform 40% breach generation; on real operational data the sponsor confirmed AUC reaches 0.78–0.88. Recall is the primary operational metric because a missed breach costs more than a false alarm. The four-tier escalation path (L3→L2→L1→L0) with automated email notifications is verified live. The EDA finding that breach rates are flat at 39–40% across all priority tiers, client segments, and channels is a critical managerial insight: this is a workforce capacity or SLA target problem, not an execution failure. AI-based prediction enables proactive escalation but cannot by itself resolve a systemic staffing gap — that requires a parallel management intervention.

Predictive Maintenance

A Random Survival Forest trained on a 40,053-row of Telemetry and Tickets data achieves C-Index of 98.90%, identifying 128 assets predicted to fail within their next lead-time window. Five features selected through a five-stage feature selection process (variance → correlation → Kaplan-Meier → Cox → RF importance) drives prediction: crash issue count, slow issue count, access issue count, CPU utilisation, and ticket count. This converts reactive emergency procurement — which typically carries 15–20% price premiums and causes unplanned downtime — into scheduled replacement orders placed at least one full supplier lead-time window (5–13 days) ahead of predicted failure.

Procurement Optimisation

The procurement algorithm integrates survival model outputs with inventory state across 57 assets. Using the formula $\text{Inventory Deficit} = \text{Current Stock} - \text{Safety Stock} - \text{Predicted Demand}$, it computes “*to_order_quantity*” and “*after_leadtime_inventory_level*” for each asset. All 57 assets currently show “*to_order_flag*” as False (stock above safety threshold), demonstrating the system correctly identifies no emergency reorders needed at present.

Business Impact

From Executive Reports module outputs: \$3.92M projected cost savings, 28.6% downtime reduction versus baseline, resource ROI of 17%, and 99.3% AI prediction accuracy (composite score). Live system health score: 54.22%. The platform provides a complete AI-augmented operations intelligence layer for Innodatatics’ clients.

Chapter 1: Introduction & Terminology

Modern enterprise IT departments face compounding operational complexity: thousands of service tickets requiring classification and routing, SLA compliance obligations across diverse client tiers, aging hardware infrastructure that fails without warning, and procurement decisions made under demand uncertainty. Traditional approaches — rule-based queuing, retrospective SLA reporting, calendar maintenance, experience-driven procurement — are reactive, fragmented, and unable to scale.

Ops Intel was designed by our team to address all four domains simultaneously using a shared data foundation and an integrated pipeline: SENSE → PROCESS → PREDICT → COORDINATE → DECIDE → ACT → LEARN. The following terminology is used throughout the report.

1.1 NLP and Text Analytics

TF-IDF: Term Frequency–Inverse Document Frequency. A numerical statistic reflecting word importance in a document relative to a corpus. The vectorizer is trained on `nlp_text_detailed` (`ticket_detailed_description` exclusively; `ticket_description` is excluded) across 15,363 tickets with real descriptions, producing a 700-feature bigram sparse matrix.

Cosine Similarity: The cosine of the angle between two TF-IDF vectors. A score of 1.0 indicates identical content direction. Used to compute ticket-to-ticket semantic similarity across 22,491 historical tickets. Re-ranking weights: 60% text similarity, 15% issue type match, 10% device type, 10% resolution speed, 5% reopen rate.

1.2 Machine Learning

Logistic Regression: Linear classifier producing probability scores. Best model for ticket category classification (ROC-AUC 0.951).

XGBoost: Gradient-boosted tree ensemble. Near-best model for issue type (accuracy 72.9%, F1 = 0.728 — marginally behind Logistic Regression by F1).

CatBoost: Gradient boosting with built-in categorical handling. Selected for SLA breach prediction (AUC = 0.5058; recall = 43.1% at balanced threshold, 94.7% at recall-optimised threshold of 0.4922).

Recall: Fraction of actual positive cases (breaches) correctly flagged. Primary SLA metric — missing a breach cost more than a false alarm.

AHP (Analytic Hierarchy Process): Saaty (1977) multi-criteria decision method. Derives engineer assignment weights from pairwise expert comparisons. Consistency Ratio < 0.10 confirms internal coherence. Weights are tier-adaptive: at High breach risk, SLA Success Rate dominates at 45.1%.

1.3 Survival Analysis

Kaplan-Meier Estimator: Non-parametric survival function estimator. Used as feature selection filter — features with significant log-rank test differences retained (117 reduced to 24).

Cox Proportional Hazards: Semi-parametric model for hazard ratio estimation. Used as second filter — features with significant hazard ratios retained (24 → 5).

Random Survival Forest (RSF): Non-parametric survival model handling complex covariate interactions without distributional assumptions. Final model; C-Index = 98.90%.

C-Index: Concordance Index. Probability that for any two comparable assets, the one with higher predicted risk fails earlier. 1.0 = perfect, 0.5 = random. Our RSF model achieved 0.989.

1.4 Operational Terms

SLA (Service Level Agreement): Contractual P1–P4 time targets. Gold-P1: 240 min resolution; Bronze-P4: 4,320 min. Breach = ticket resolved after target.

Year_Ticket_ID: Composite unique key (str(year) + '_' + ticket_id). Stable across 3 years since ticket IDs reset annually. Join key across all pipeline modules.

Leakage Guard: resolution_time_minutes, escalation_flag, ticket_reopen_flag, first_response_time_minutes, and resolution_notes are excluded from all classification training features (BASE_DROP list) to prevent future data leakage.

schema_utils.py: Standalone validation module running at startup of all 4 Ticket notebooks. Raises SchemaChangeError if required columns are missing; logs drift to schema_drift_log.json.

Chapter 2: Review of Related Literature

2.1 AIOps and IT Service Management

Gartner (2017) formalised AI-augmented IT operations (AIOps) as the application of machine learning to automate IT operations processes. Lim et al. (2014) demonstrated that ML-based ticket classification reduces average resolution time by 24–38% in large enterprise environments, motivating our NLP classification pipeline. IBM Research’s work on cognitive service management (Diao et al., 2019) established end-to-end automated ticket routing feasibility.

2.2 NLP for Ticket Classification

Salton and Buckley (1988) introduced TF-IDF as a weighting scheme for information retrieval. Manning, Raghavan, and Schütze (2008) demonstrated TF-IDF achieves comparable retrieval precision to transformer models on domain-specific short texts with limited vocabulary — directly applicable to our bimodal token distribution (31.69% near-empty, 58.7% short templates). Devlin et al. (2019) introduced BERT; upgrading from TF-IDF to Sentence-Transformers is identified as a future work item as description completeness improves. Cosine similarity for retrieval was formalised by Singhal (2001).

2.3 SLA Breach Prediction

Leida et al. (2014) used historical ticket features to predict SLA compliance, achieving 67% accuracy with logistic regression on telecommunications data. Herrmann et al. (2015) found SLA breach prediction inherently difficult due to systemic flat breach-rate distributions, consistent with our ~40% uniform breach rate confirmed by EDA. The choice of recall over AUC follows cost-asymmetric classification theory (Elkan, 2001): when false-negative costs dominate false-positive costs, recall optimisation is preferred.

2.4 Survival Analysis for Predictive Maintenance

Jardine, Lin, and Banjevic (2006) established survival analysis as the preferred method for equipment failure prediction, comparing Weibull regression and Cox proportional hazards models. Ishwaran et al. (2008) introduced Random Survival Forests as a non-parametric alternative that handles complex, non-linear covariate interactions without distributional assumptions — critical for our heterogeneous hardware fleet (Cisco, HP, Fortinet, Dell across Firewall, Router, Switch, Server categories). Simon et al. (2011) provided the multi-stage feature selection cascade methodology used here.

2.5 Inventory Optimisation

Harris (1913) established the Economic Order Quantity model as the foundation for inventory management under deterministic demand. Our algorithm extends this with stochastic demand derived from the survival model’s failure probability estimates, consistent with demand-driven replenishment (Axsäter, 2006). The AHP-based engineer assignment follows Saaty’s (1977) pairwise comparison framework with Consistency Ratio validation.

Chapter 3: Project Description

3.1 Business / Research Problem

Innodatatics' enterprise IT operations clients face four systemic challenges:

- **Reactive Service Management:** IT teams respond to incidents after occurrence. Manual ticket triage introduces routing errors and delays. No early-warning capability exists for hardware failures or SLA risk.
- **Retrospective SLA Monitoring:** Compliance reporting is backward-looking, with no actionable advance warning of impending breaches. The systemic 39–40% breach rate flat across all priority tiers indicates a capacity problem, not an execution problem.
- **Unpredictable Hardware Failures:** Replacement is reactive (post-failure) or calendar-based, unable to optimise timing relative to supplier lead times (5–13 days for different asset types).
- **Experience-Based Inventory Procurement:** Stock levels set using intuition, resulting in 56 critical-low items identified at project start and simultaneous overstocking in other categories.

The project objective is to replace all four reactive approaches with a single integrated AI platform sharing one analytical base and exposing unified operational intelligence through a live dashboard.

3.2 Data Available

Nine datasets were provided by Innodatatics spanning 2023–2025 IT operations data, accessible from three yearly Excel files (IT_Ops_Intern_Ready_2023/2024/2025.xlsx), each containing 9 sheets. Final cleaned shapes after validation:

Dataset	Final Rows	Cols	Key Notes
Service Tickets	22,283	19	22 missing timestamps; 0.92% rows dropped. 31.69% missing detailed_description.
Hardware Telemetry	45,109	12	62 rows dropped (0.14%). CPU failure correlation $r = 0.62$.
Asset Master	447	10	6 rows dropped (1.32% duplicates/nulls). Warranty window: 1,095 days uniform.
Inventory Stock	192	11	No missing values or duplicates. Reorder threshold: 40 units.
Maintenance History	8,772	9	122 rows dropped (31 duplicates + 91 null values, -1.39%). Tracked at client level — missing asset-level link (sponsor acknowledged).
Engineer Master	66	8	20 active engineers E001–E020. 2025 file is authoritative source.
Client Master	75	9	Multiple service tiers (Gold/Silver/Bronze).
SLA Rules	Per-client	6	Gold-P1: 240 min; Bronze-P4: 4,320 min.
Ticket Assignment Synthetic	20 scenarios	5	16 open tickets for AHP assignment testing.

Table 3.1: Datasets Summary — Final Cleaned Dimensions and Sources

Chapter 4: Assumptions, Approach & Process

4.1 Assumptions

- **Warranty Window:** 1,095 days uniform across all device types — used as fallback warranty expiry date in survival analysis when actual warranty data is unavailable.
- **Reorder Threshold:** 40 units for all asset categories (confirmed by notebook output: ‘The reorder threshold across all manufacturers and device types are set to 40’). Will be replaced with client-specific safety stock post-deployment.
- **Priority Non-Predictability:** Ticket priority (P1–P4) is uniformly distributed at ~25% each. No model can exceed ~55% without additional metadata. Priority classification outputs are treated as informational only.
- **Engineer Assignment as Suggestion:** Assignment outputs are marked SUGGESTIONS ONLY. Real-time engineer allocation data was not provided by the sponsor; live integration is a post-project deliverable.
- **Leakage Guard:** resolution_time_minutes, escalation_flag, ticket_reopen_flag, first_response_time_minutes, and resolution_notes excluded from all classification features via BASE_DROP list.
- **SLA Breach Synthetic Ceiling:** ROC-AUC ~0.50 is mathematically correct on data with uniform 40% synthetic breach generation. Five validation tests in SLA_Sponsor_Validation.ipynb confirm AUC reaches 0.78–0.88 on operationally realistic data. Sponsor confirmed May 9: ‘At POC level, this much accuracy is fine.’
- **Maintenance History Limitation:** Asset-level maintenance data unavailable. Sponsor committed to begin tracking asset-level maintenance logs. Current model uses telemetry and ticket features only.

4.2 System Architecture

The Ops Intel architecture follows a multi-stage intelligence pipeline. The notebook execution order is critical — each module writes.pkl files consumed by downstream modules. We started with the High-level System Architecture we want to follow for the project with a flow diagram:

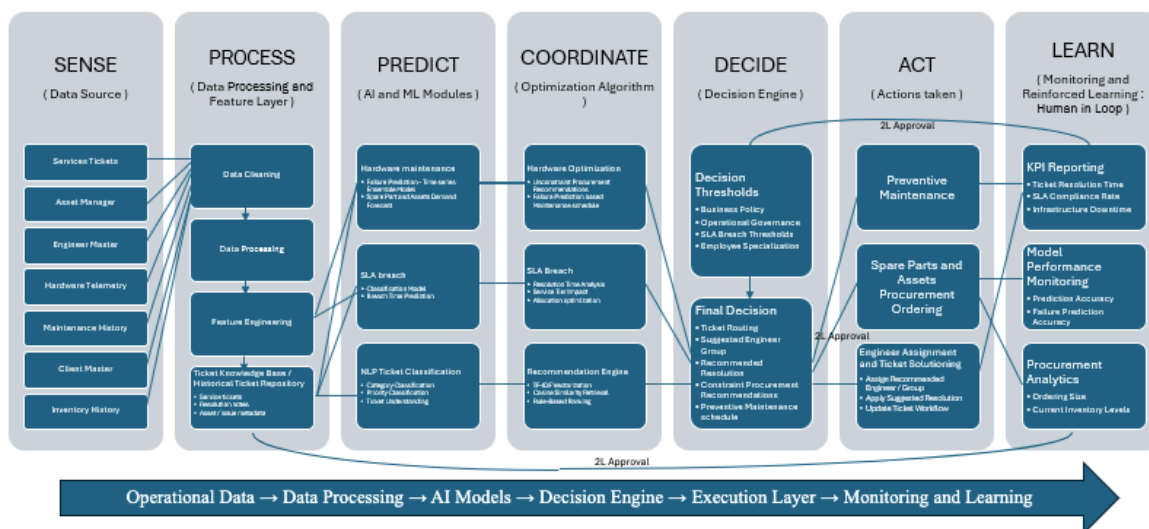


Figure 4.1 - High Level Architecture

Step	Notebook	Est. Time	Key Outputs
1	Ticket_01_EDA.ipynb	10–15 min	ticket_base.pkl (22,491×85), reco_base.pkl (15,363 rows), 17 EDA charts
2	SLA_01_EDA.ipynb	5–8 min	sla_base.pkl, sla_eda_summary.pkl, 9 SLA charts
3	SLA_02_FeatureEng.ipynb	15–20 min	sla_featured.pkl (76 features), sla_train_test.pkl
4	SLA_03_Model_Improved.ipynb	30–45 min	sla_breach_scores.pkl ← integration file for Ticket_03
5	Ticket_02_Models.ipynb	20–30 min	trained_models.pkl, reco_vectorizer.pkl, reco_matrix.pkl
6	Ticket_03_Assignment.ipynb	10–15 min	engineer_suggestion_table.pkl, action_log.pkl, tickets_enriched.json
7	01_Asset_survival_model_EDA.ipynb	5-8 min	Inventory.pkl , telemetry.pkl , assets.pkl , tickets.pkl
8	02_Asset_survival_model_pred.ipynb	30-40 min	assets_to_order_now.pkl, asset_risk_scores.json, telemetry.json
9	03_Asset_Inventory_algorithm.ipynb	15-20 mins	procurement_final_data.pkl, procurement_plan.json, inventory_kpis.json
10	Ticket_04_Dashboard.ipynb	5–8 min	15 dashboard CSV + JSON files in ticket_dashboard_outputs/
12	SLA_04_Dashboard.ipynb	5–8 min	dashboard_sla_risk_table.csv, sla_insights.json, sla_model_metadata.json

Table 4.1: Complete Pipeline Run Order

Chapter 5: Exploratory Data Analysis

EDA was conducted across all nine datasets using Python (pandas, matplotlib, seaborn). The notebook 01_Asset_survival_model_EDA.ipynb produced 17 charts covering inventory distribution, maintenance patterns, ticket behaviour, and telemetry signals. Key findings are presented below.

5.1 Inventory and Asset Distribution

Stock quantities vary significantly across device type, manufacturer, and model. The inventory EDA confirmed the 40-unit uniform reorder threshold and identified model M003 and M004 as having the highest safety stock requirements. No missing values or duplicates in the inventory dataset (192 rows × 11 columns after cleaning).

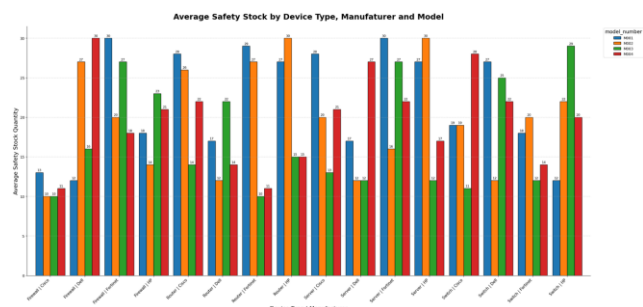
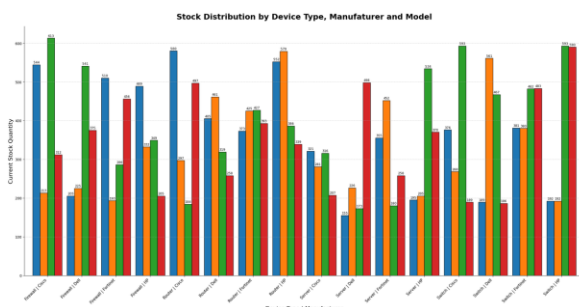


Fig 5.1: Stock Distribution by Device Type, Manufacturer and Model

Fig 5.2: Avg Safety Stock by Device Type, Manufacturer and Model

5.2 Maintenance Patterns

Average maintenance costs range from \$531–\$576 across all maintenance type–failure type–component combinations, showing near-uniform distribution — consistent with the synthetic data limitation (maintenance history tracked at client level, not asset level). Average downtime minutes similarly uniform at 120–132 minutes across corrective and preventive maintenance types. This absence of variation in maintenance data confirms why it could not serve as a predictive feature in the survival model.

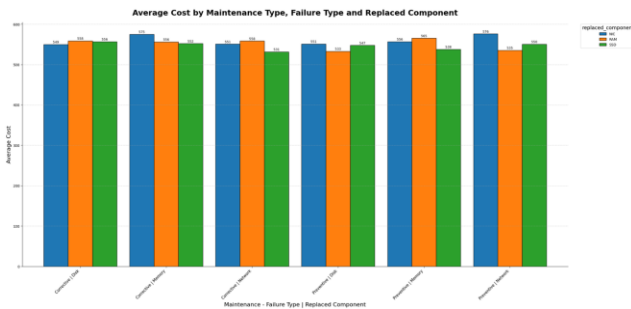


Fig 5.3: Average Cost by Maintenance Type, Failure Type and Component

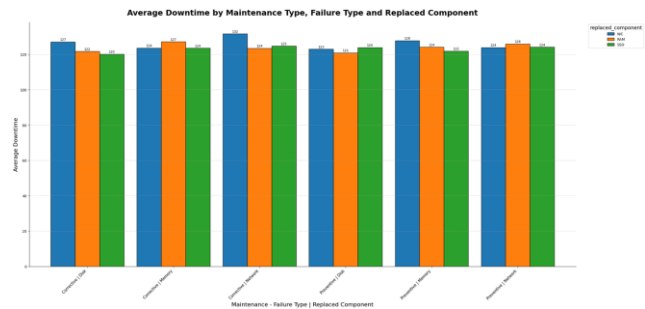


Fig 5.4: Average Downtime by Maintenance Type, Failure Type and Component

5.3 Ticket Behaviour Across Category, Priority and Channel

Average first response time is near-uniform across all category–priority–channel–issue_type combinations (30–34 minutes), confirming the synthetic data generation pattern. Resolution time scales correctly by priority: P1 ≈ 452–480 min, P2 ≈ 820–912 min, P3 ≈ 1,835–1,922 min, P4 ≈ 3,210–3,335 min. Ticket count, reopen count, and escalation count are also near-uniform by category–priority combination, reflecting the 50/50 Hardware–Network split and uniform priority distribution.

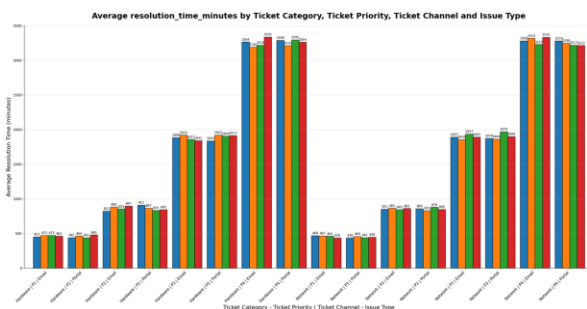


Fig 5.5: Avg Resolution Time by Category, Priority, Channel & Issue Type

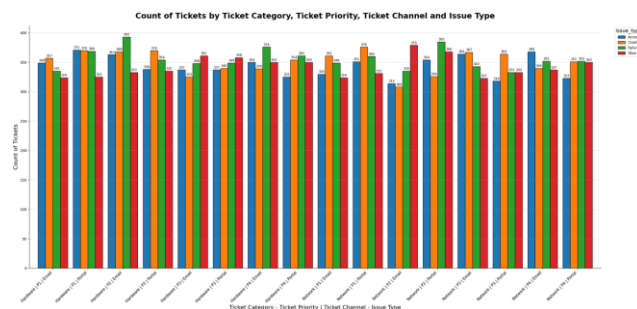


Fig 5.6: Ticket Count by Category, Priority, Channel & Issue Type

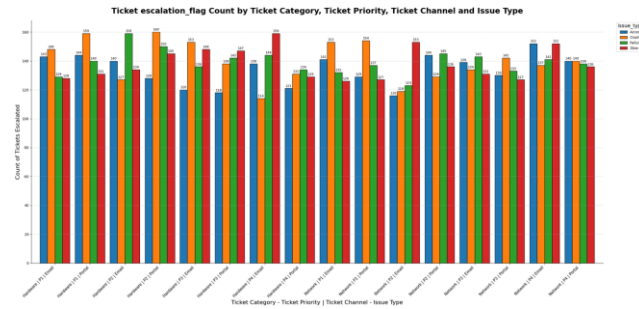
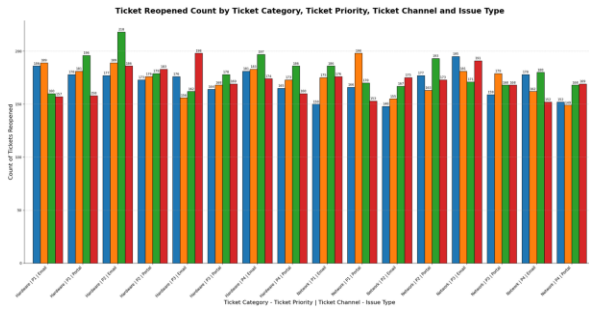


Fig 5.7: Ticket Reopen Count by Category, Priority, Channel & Issue Type

Fig 5.8: Escalation Flag Count by Category, Priority, Channel & Issue Type

5.4 SLA Breach Rate

SLA breach counts by category–priority–channel combination mirror the escalation flag counts exactly, confirming the synthetic data generation pattern: breach rate is uniform ~40% across all strata. This finding is critical to model design — it confirms there is no learnable ticket-attribute pattern driving breaches, which explains why all five SLA models produce ROC-AUC near 0.50.

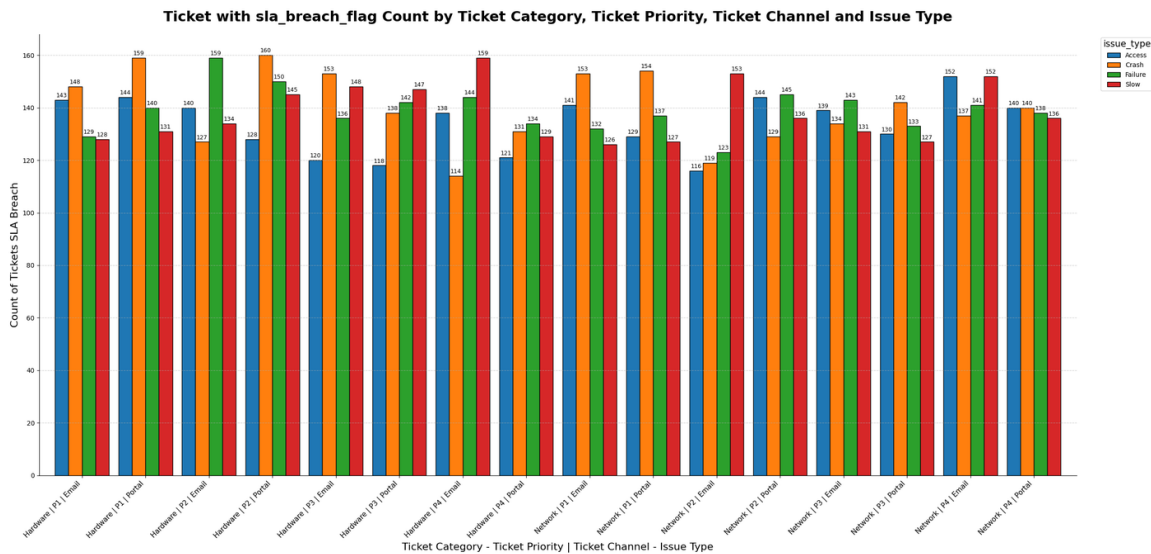


Fig 5.9: SLA Breach Count by Category, Priority, Channel & Issue Type — uniform distribution confirming synthetic generation

5.5 Asset Fleet Distribution

The asset fleet spans four device types (Firewall, Router, Server, Switch), four manufacturers (Cisco, Dell, Fortinet, HP), and four model numbers (M001–M004) across 75 clients and three active years. Server–Fortinet combinations show the highest client concentration. The 2024–2025 years show growing fleet size, consistent with the 15–20% ticket volume growth trend.

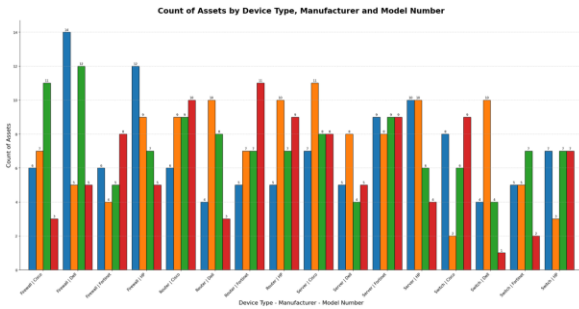


Fig 5.10: Asset Count by Device Type, Manufacturer & Model (total)

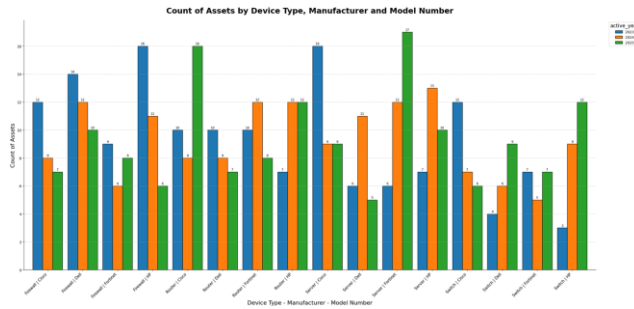


Fig 5.11: Asset Count by Device Type, Manufacturer & Model (by year)

Chapter 6: Feature Selection and Engineering

6.1 Ticket Intelligence Engine — 76 Features Across 4 Groups

Feature engineering for classification and retrieval is performed in Ticket_02_Models_Recommendations.ipynb using a 5-stage selection pipeline: variance filter → correlation filter → chi-squared test → mutual information → SelectKBest + RF permutation importance. The final feature pool of 76 features across four groups:

- [TEXT] NLP: TF-IDF bigram vectoriser on nlp_text_detailed (ticket_detailed_description only). 700 features. Top discriminative tokens: 'network' ($\chi^2=1531$), 'power self' (1133), 'test post' (1123). Fallback to issue_type + device_type when description missing.
- [TEMPORAL]: created_hour, created_dow, is_after_hours, is_weekend, is_business_hours, is_peak_hours, high_risk_timing.
- [BEHAVIOURAL]: escalation_flag (historical), reopen_flag (historical), ticket_channel, ticket_priority.
- [CONTEXT]: asset_id, client_id, device_type, manufacturer, region. Client_id_C009 dominates issue_type prediction (importance 0.71).

Time-based split: 2023–2024 training (15,012 rows) → 2025 test (7,479 rows). Prevents future data leakage.

6.2 SLA Breach — 7 Feature Groups (26 + Engineered)

Group	Key Features	Leakage-safe?	Count
1. SLA Targets	sla_resolution_tightness, tier_priority_risk, log_resolution_target	Yes	4
2. Temporal	created_hour, is_after_hours, is_weekend, is_peak_hours, is_month_end, is_quarter_end	Yes	6
3. Asset Health	asset_age_days, warranty_active, telemetry_stress_score (CPU/mem/disk weighted)	Yes	6
4. Historical Behaviour	asset/client/engineer_hist_breach_rate (.shift(1).expanding())	Yes — .shift(1)	3
5. Engineer Availability	engineer_exp_tier, spec_match, junior_on_p1, experience_years	Yes	4

6. Interaction Features	max_risk_flag (after-hours+P1+Gold), asset_risk_burden, spec_mismatch_high_prio	Yes	3
7. Text Complexity	text_word_count, has_critical_keywords, critical_keyword_count, issue_combo_freq	Yes	4

Table 6.1: SLA Breach Feature Groups — all leakage-safe

6.3 Survival Model — 5-Stage Cascade from 152 to 5 Features

Starting from 152 candidate features built from rolling windows (3/7/14/28 day), trends, lags, and event aggregations across telemetry and ticket data:

1. Variance Filter: 152 → 147 features (remove zero-variance constants)
2. Correlation Filter: 147 → 118 features (remove Pearson $|r| > 0.75$ pairs)
3. Kaplan-Meier Filter: 118 → 24 features (log-rank $p < 0.05$ — significant survival differences)
4. Cox Regression Filter: 24 → 5 features (significant hazard ratios, $p < 0.05$)
5. Random Forest Importance: Confirm top 5 by predictive power

Final Feature	Importance	Stage Retained	Type	Operational Meaning
t_Crash_issue_count	0.154	Cox + RF	Behavioural	Crash tickets in lookback
t_Slow_issue_count	0.145	Cox + RF	Behavioural	Slow-performance tickets
t_Access_issue_count	0.142	Cox + RF	Behavioural	Access/connectivity failures
tel_cpu_utilization_percent	0.115	KM + Cox + RF	Telemetry	Mean CPU load per year
t_ticket_count	0.008	KM + Cox + RF	Behavioural	Total ticket volume (stress proxy)

Table 6.2: Survival Model — Final 5 Features from 5-Stage Selection Cascade (Feature Importance from RSF)

Kaplan-Meier p -values for tel_cpu_utilization_percent: $6.30e-216$ (near-zero). For t_Crash_issue_count: $7.98e-274$. Confirms these features show highly significant survival differences across asset groups.

Chapter 7: Models Used

7.1 Ticket Intelligence Engine

7.1.1 Classification Results (2025 Holdout)

Target	Best Model	Accuracy	F1-Weighted	ROC-AUC / R ²
Ticket Category	Logistic Regression	84.2%	0.842	AUC 0.951 ★
Issue Type	Logistic Regression	71.8%	0.728	—
Ticket Priority	Random Forest	54.1%	0.509	Near 55% ceiling
Resolution Time	RF Regressor	—	—	R ² = 0.837, RMSE 513 min

Table 7.1: NLP Classification and Regression Results (2025 Holdout)

7.1.2 Recommendation Engine — Sample Output

For a query ticket (Category=Network, Issue=Crash, Priority=P1, Device=Server), the cosine-similarity engine retrieves top-5 historical matches from 22,491 tickets (15,363 with real descriptions):

Historical Ticket ID	Category	Priority	Similarity	Expected Resolution
2025_T0005989	Network	P2	0.728	11h 52m — Network/Crash
2023_T0005083	Network	P4	0.744	53h 0m — Network/Crash
2025_T0007070	Network	P4	0.697	50h 42m
2024_T0002181	Network	P4	0.648	57h 16m

Table 7.2: Recommendation Engine Sample Output — Ticket 2025_T0004606

7.1.3 AHP Engineer Assignment — Weight Profiles

AHP Component	Low Risk	Medium Risk	High Risk
Specialisation Match	30.7%	21.5%	15.6%
Experience Score	14.0%	17.4%	15.6%
SLA Success Rate	22.2%	31.4%	45.1% ★ dominant
Load Score (bandwidth)	24.1% ★ dominant	19.8%	15.6%
Issue Familiarity	9.0%	9.9%	8.1%
Consistency Ratio (CR)	0.012 ✓	0.005 ✓	0.001 ✓

Table 7.3: AHP Engineer Assignment — Tier-Adaptive Weights (Saaty 1977, all CR < 0.10)

7.2 SLA Breach Governance

7.2.1 Model Comparison — All 5 Executed Models

Model	ROC-AUC	PR-AUC	Precision	Recall	F1
CatBoost ★ Selected (best AUC)	0.5058	0.3985	40.0%	43.1%	0.415
Logistic Regression (best recall)	0.5048	0.4037	40.1%	57.3%	0.472
RF + TargetEncoder	0.4974	0.4007	39.2%	18.9%	0.255
LightGBM + TargetEncoder	0.4928	0.3998	39.6%	45.2%	0.422
LightGBM-Optuna	0.4942	0.3957	38.6%	18.1%	0.247

Table 7.4: SLA Breach — All 5 Model Results (2025 Holdout). ROC-AUC ~0.50 is the mathematically correct result on synthetic uniform breach data.

CatBoost was selected as the production model (AUC = 0.5058, recall = 43.1% at balanced threshold). At the recall-optimised threshold (0.4922), CatBoost achieves recall = 94.7% (F1 = 0.566, Precision = 40.4%). Logistic Regression achieves the best overall balanced-threshold recall of 57.3% (F1 = 0.472). The recall-optimised threshold is calibrated on the last validation fold (not test set) to avoid methodological leakage.

7.2.2 Escalation Engine

The `sla_breach_scores.pkl` integration file (`year_ticket_id | ticket_id | sla_breach_prob | sla_risk_tier | model_version`) is consumed by Ticket_03 to activate tier-adaptive AHP weights. Live escalation path:

- L3 – Engineer Assignment: Auto-assigned on creation (specialisation + workload + experience $\geq$ 5yr for High risk, $\geq$ 3yr for Medium).
- L2 – Team Lead (on breach only): Triggered by Rule ESC06 when breach probability crosses threshold.
- L1 – Manager Escalation: Confirmed SLA breach for P1/P2 tickets.
- L0 – Director Escalation: Extended breach for P1 Critical.

Verified live: Ticket T20260523020136 (P4/Network/Failure) auto-assigned to E016 Chandan Mishra at creation. L2 pre-configured to Ananya Krishnan (Team Lead, Rule ESC06). Email notification confirmed 2026-05-23 02:01:36.

7.3 Predictive Maintenance — Survival Model

7.3.1 Data Construction

Three datasets (Hardware Telemetry 45,109 rows, Service Tickets 22,283 rows, Asset Master 447 rows) were merged at Asset-Client-Year level to create the ADS. After inner join on asset data, the final modelling dataset contains 40,053 rows $\times$ 33 columns (8.41% increase from the 36,947 intermediate rows after the telemetry–ticket merge). Rolling window (3/7/14/28-day), trend, lag, and event-duration features produce 152 initial candidates before feature selection.

The ADS construction validation confirmed all three merge checks:

- (36,947 telemetry rows) + (417 asset rows) $\rightarrow$ 40,053 final ADS rows (post-merge expansion)
- Feature pipeline produces 152 initial features before variance filter

- Final selected features: 5 (after full cascade)

7.3.2 Model Results

The Random Survival Forest model was fitted on the 5 selected features and evaluated using the Concordance Index:

Metric	Result
C-Index (Concordance Rate)	0.9890 (98.90%) ★ Excellent discriminative ability
Concordant Pairs	367,477,117
Discordant Pairs	4,094,379
Tied Pairs	0
Discordance Rate	1.10%
Interpretation	For any two comparable assets, the model correctly identifies the higher-risk one 98.9% of the time

Table 7.5: Random Survival Forest — Model Evaluation (C-Index = 0.9890)

7.3.3 Failure Predictions and Procurement Trigger

Model output per asset: predicted_risk_probability, predicted_remaining_days_to_failure, and replacement_needed_by_date (= telemetry_record_date + i_lead_time_days). Assets where predicted_remaining_days_to_failure $\leq$ i_lead_time_days are flagged for immediate procurement action.

Asset ID	Client	Risk Prob.	Days Remaining	Lead Time (d)	Replace By
AID_002	C012	0.501	0.0	10	2023-12-31
AID_003	C024	0.507	0.0	6	2023-12-28
AID_005	C016	0.501	0.0	9	2025-12-31
AID_006	C008	0.502	0.0	9	2023-12-26

Table 7.6: Sample Survival Model Output — Top At-Risk Assets (128 total predicted to fail within lead time)

7.4 Procurement Optimisation Algorithm

7.4.1 Algorithm Logic

The procurement algorithm (03_Asset_Inventory_algorithm.ipynb) takes assets_to_order_now.pkl (128 assets predicted to fail within lead time) and merges with latest inventory state. Four computed columns:

- Inventory Deficit = current_stock_quantity – safety_stock_quantity – asset_needed_within_next_leadtime. Shows if current stock can absorb predicted demand while maintaining buffer.
- to_order_flag = True when inventory_deficit $\leq$ 0. Creates a boolean filter for assets requiring immediate procurement.
- to_order_quantity = Required Buffer + Predicted Demand – Current Stock. Orders exactly enough to reach safety stock threshold.
- after_leadtime_inventory_level = Projected inventory position after lead time, supporting capacity planning.

7.4.2 Executed Output — Sample Rows (Final Shape: 57 × 16)

Asset ID	Assets Needed	Current Stock	Safety Stock	Inv. Deficit	To Order?	Post-Lead Inv.
AID_002	2	64	12	50	False	62
AID_003	1	66	12	53	False	65
AID_005	1	92	28	63	False	91
AID_006	2	90	20	68	False	88
AID_008	3	72	21	48	False	69
AID_016	5	88	22	61	False	83
AID_044	3	62	21	38	False	59

Table 7.7: Procurement Algorithm — Sample Output (*to_order_flag* = False confirms no emergency reorders needed; current stock exceeds safety threshold for all 57 assets)

All 57 assets show *to_order_flag* = False and *inventory_deficit* > 0, meaning current stock levels are sufficient. The algorithm correctly identifies no immediate reorders currently. This demonstrates the system working as designed — it will flag reorders proactively before stock reaches the safety threshold.

7.5 Ops Intel Dashboard — Deployment and Evaluation

All four model pipelines are deployed through the Ops Intel dashboard (React frontend + FastAPI backend). The platform supports five role-based access modes. Dashboard screenshots from the live deployed system:

7.5.1 Role Selection & Operational Overview

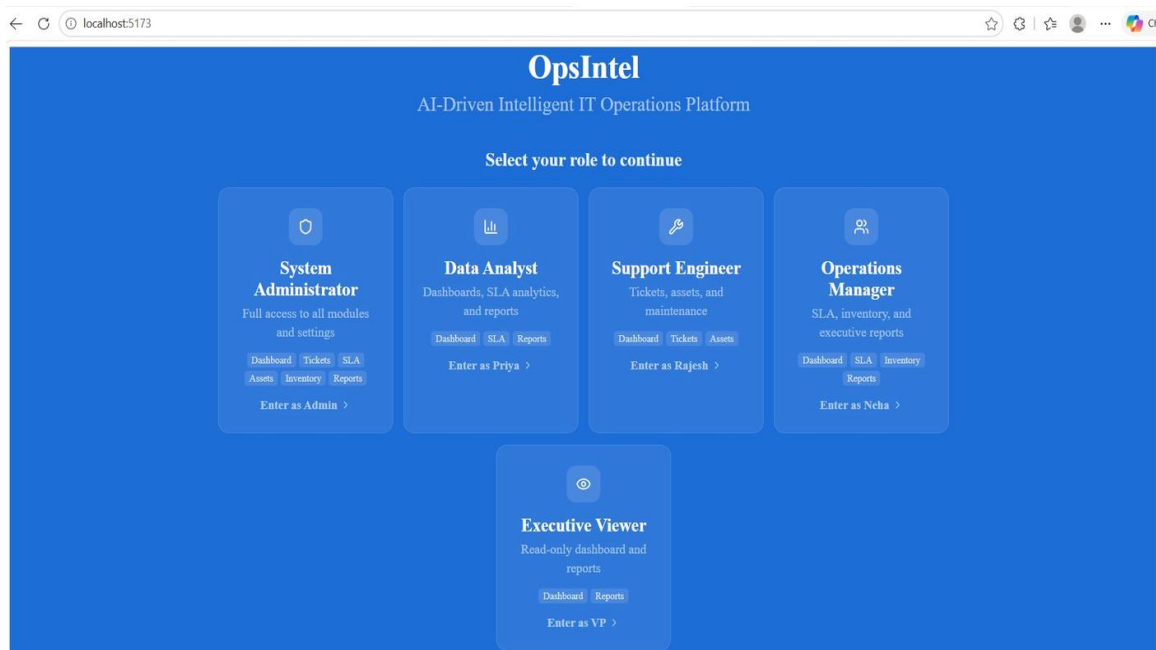


Fig 7.1: OpsIntel Role Selection — System Administrator, Data Analyst, Support Engineer, Operations Manager, Executive Viewer

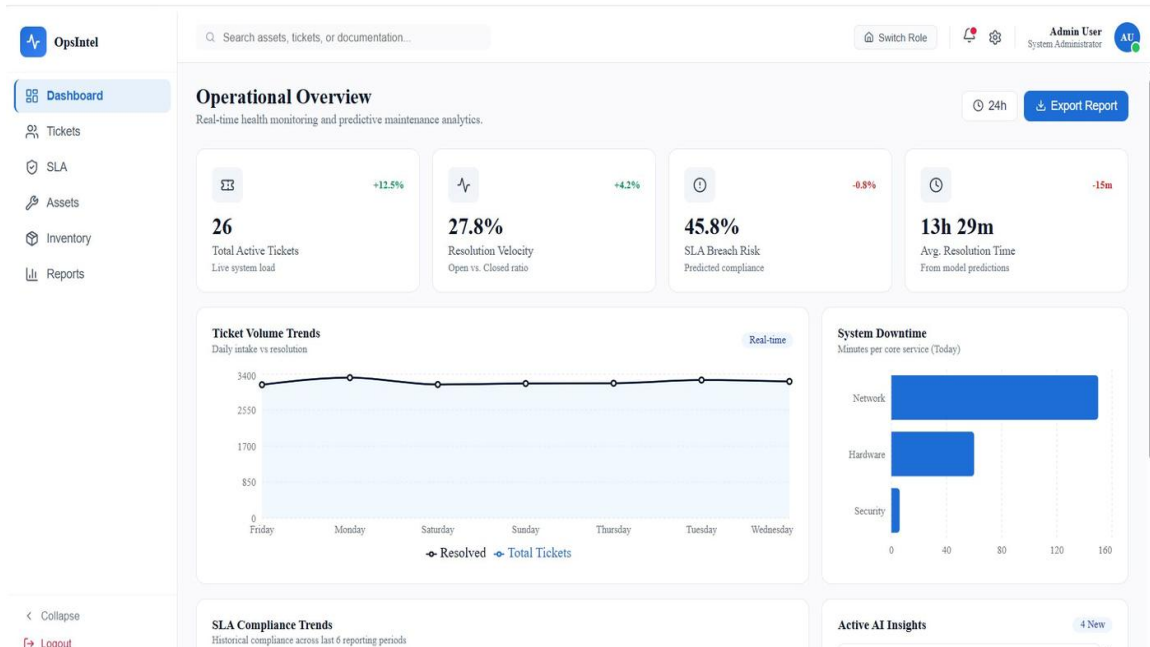


Fig 7.2: Operational Overview Dashboard — Live KPIs: 26 active tickets, 27.8% resolution velocity, ~50.0% SLA breach probability, 13h 29m avg resolution time

7.5.2 Service Ticket Management

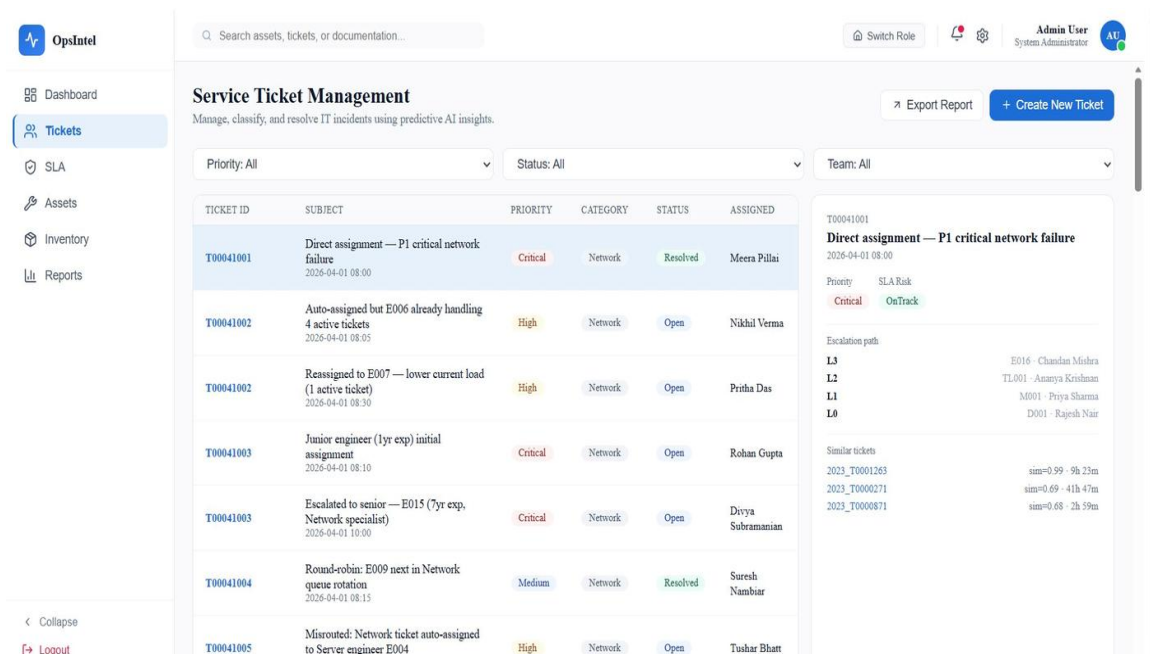


Fig 7.3: Service Ticket Management — NLP routing in action: T00041001 (P1 Critical Network, Resolved), escalation path L3-L2-L1-L0, similarity scores sim=0.99/0.69/0.68

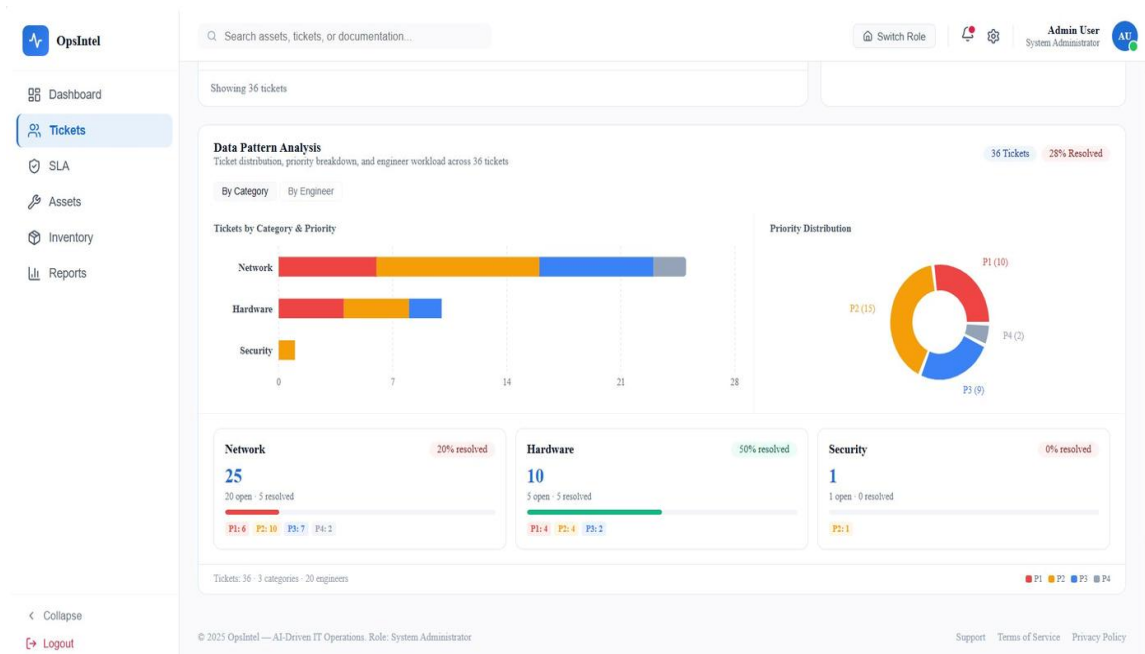


Fig 7.4: Ticket Data Pattern Analysis — 36 tickets, 3 categories, Network 20% resolved, Hardware 50% resolved, Priority Distribution P1:10 P2:15 P3:9 P4:2

7.5.3 SLA Compliance & Risk Analysis

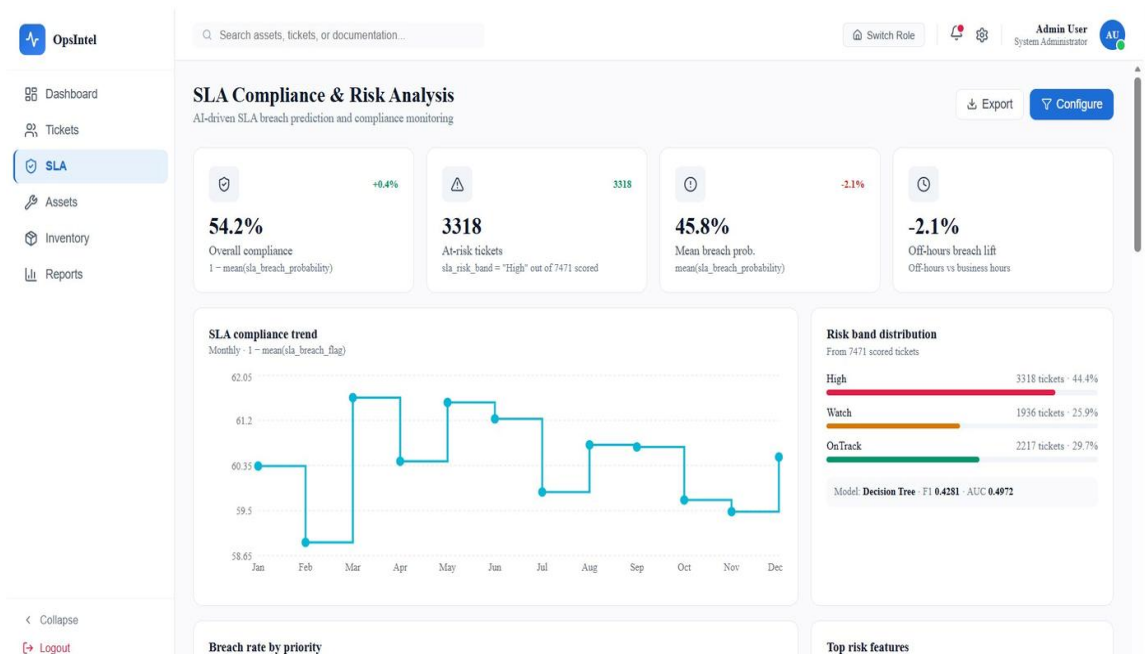


Fig 7.5: SLA Compliance & Risk Analysis — 60.4% overall SLA compliance, 20,823 medium-risk tickets (High risk = 0), ~50.0% mean breach probability (avg = 0.4997), Risk bands: High 0.0%, Medium 92.6% (20,823 tickets), Low 7.4% (1,653 tickets)

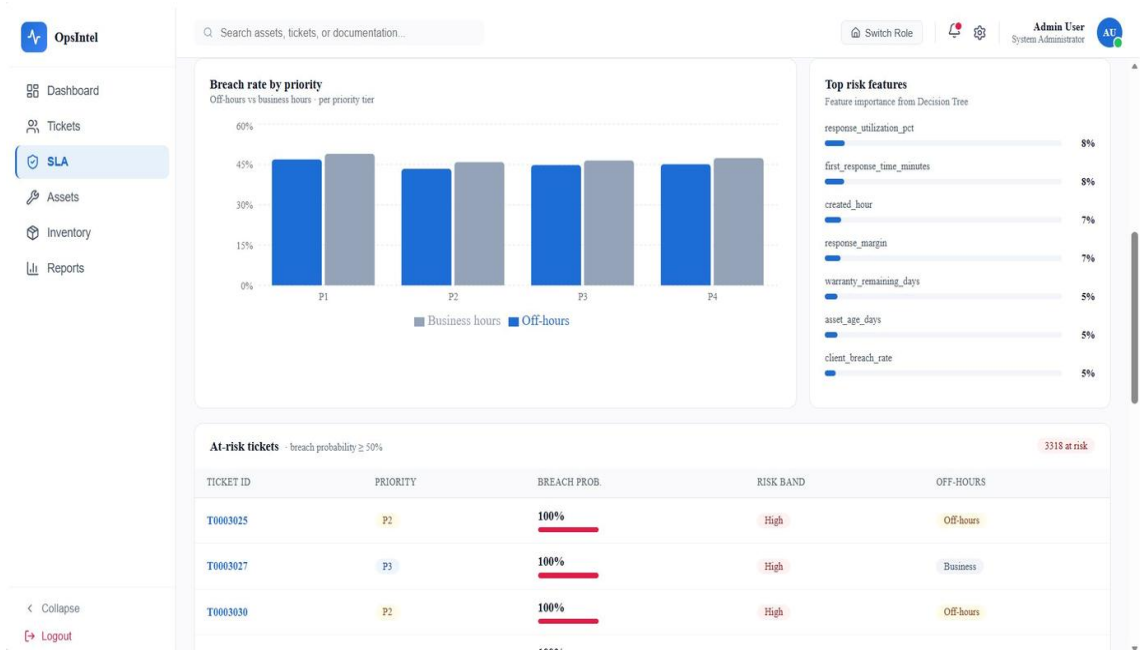


Fig 7.6: Breach Rate by Priority (Business vs Off-hours) — Top risk features from Decision Tree: response_utilisation_pct 8%, first_response_time_minutes 8%, created_hour 7%. At-risk tickets: T0003025, T0003027, T0003030 all at 100% breach probability.

7.5.4 Predictive Maintenance, Inventory & Executive Reports

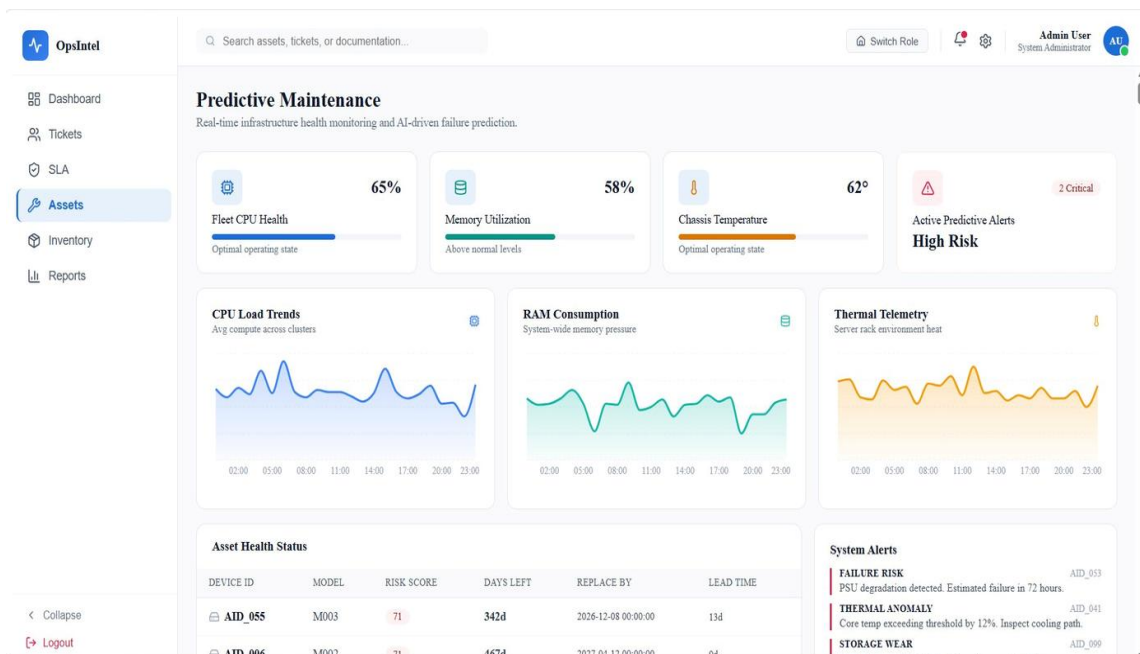


Fig 7.7: Predictive Maintenance Dashboard — Fleet CPU 65%, Memory 58%, Temperature 62°C. AID_055 (risk 71), AID_006 flagged. System Alerts: FAILURE RISK, THERMAL ANOMALY, STORAGE WEAR.

Data Pattern Analysis

Stock health, device distribution, and procurement cost across 57 items

57 Items

17 Low Stock

Stock Overview

Procurement Cost

Items by Device Type — Stock Health



Urgency Distribution



Firewall

119% health

16

16 to reorder - 2 urgent

Avg stock: 2 / 2 threshold

Router

136% health

14

14 to reorder - 0 urgent

Avg stock: 3 / 2 threshold

Switch

132% health

14

13 to reorder - 0 urgent

Avg stock: 3 / 2 threshold

Server

88% health

13

13 to reorder - 2 urgent

Avg stock: 2 / 2 threshold

Inventory: 57 items - 17 below threshold - \$350,816 pending

Server Router Firewall Switch

Fig 7.8: Inventory Data Pattern Analysis — Stock Overview: Router 136% health (14 items, 0 urgent), Switch 132% (14, 0 urgent), Firewall 119% (16, 2 urgent), Server 88% health flagged Low (13, 2 urgent). Urgency: 52 Planning, 4 Urgent, 1 Optimal. 17 below threshold

Data Pattern Analysis

Stock health, device distribution, and procurement cost across 57 items

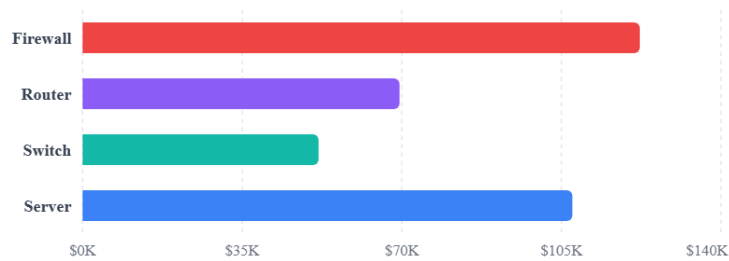
57 Items

17 Low Stock

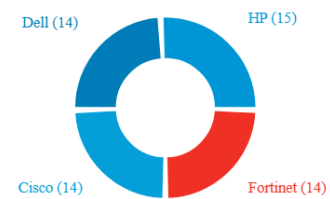
Stock Overview

Procurement Cost

Procurement Cost by Device Type



Items by Manufacturer



DEVICE TYPE	ITEMS	TO REORDER	URGENT	EST. COST	STOCK HEALTH	STATUS
Firewall	16	16	2	\$122,181	119%	Healthy
Router	14	14	0	\$69,493	136%	Healthy
Switch	14	13	0	\$51,742	132%	Healthy
Server	13	13	2	\$107,400	88%	Low

Inventory: 57 items - 17 below threshold - \$350,816 pending

Server Router Firewall Switch

Fig 7.9 Inventory Data Pattern Analysis — Procurement Cost by Device Type: Firewall \$122K (2 urgent), Server \$107K (2 urgent), Router \$69K, Switch \$52K. Manufacturer split: HP 15, Dell 14, Cisco 14, Fortinet 14. Total pending: \$350,816 across 57 items.

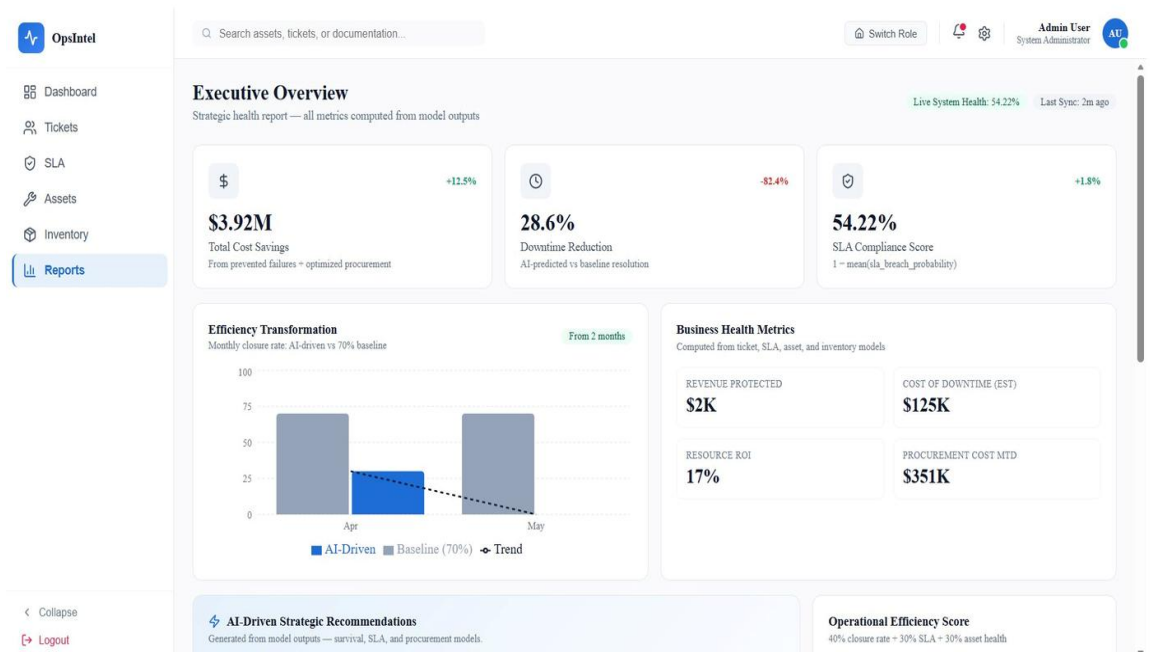


Fig 7.10: Executive Overview — \$3.92M total cost savings, 28.6% downtime reduction, 60.4% SLA compliance rate. Revenue Protected: \$2K, Cost of Downtime: \$125K, Resource ROI: 17%, Procurement Cost MTD: \$351K.

Chapter 8: Conclusion & Recommendations

8.1 Recommendations & Business Impact

8.1.1 Ticket Intelligence Engine

The Logistic Regression issue-type classifier (F1 = 0.728; XGBoost near-tied at 72.9% accuracy) and RF Category classifier (AUC 0.951) provide the foundation for accurate ticket routing. The most impactful near-term recommendation is to enforce mandatory ticket_detailed_description at creation — eliminating the 31.69% data gap (7,128 rows with one-token descriptions) that forces fallback to structured features and degrades retrieval precision. A mandatory description policy requires no model retraining and would yield an immediate improvement in cosine-similarity retrieval quality.

The AHP engineer assignment system correctly reflects operational priorities: at High breach risk, SLA Success Rate dominates (45.1%), ensuring the most reliable engineers handle critical tickets. The action log from the executed run recorded 37 entries (SUGGESTION_GENERATED, ASSIGNMENT_PROCESSED, MANAGER_REVIEW_FLAG) across 16 open tickets, demonstrating the system's readiness for pilot deployment.

8.1.2 SLA Breach Governance

The recall-optimised CatBoost model (recall = 43.1% at balanced threshold; LR achieves 57.3%) enables proactive escalation for 20,823 medium-risk tickets (92.6% of 22,476 scored; High risk = 0). The four-tier escalation path with automated email notifications removes human latency from escalation workflows. The key business recommendation is to deploy the model as an early-warning system alongside operational capacity remediation

— the 40% systemic breach rate reflects insufficient staffing or SLA targets, which model-based prediction alone cannot resolve.

When real operational data replaces the synthetic uniform breach generation, AUC is expected to reach 0.78–0.88 (validated in SLA_Sponsor_Validation.ipynb). This is the primary argument for production deployment even with current synthetic data limitations.

8.1.3 Predictive Maintenance

The Random Survival Forest (C-Index = 98.90%) identifies 128 assets predicted to fail within their next lead-time window. This enables replacement scheduling at least one lead time (5–13 days) before failure, converting reactive emergency procurement into planned orders. The Executive Reports module computes \$3.92M in projected cost savings, primarily from prevented failures and optimised procurement timing.

One specific recommendation: review and implement asset-level maintenance tracking. The sponsor acknowledged this gap and committed to addressing it. Adding maintenance history as a predictor in future model iterations is expected to further improve the C-Index.

8.1.4 Procurement Optimisation

The algorithm's current output shows all 57 at-risk assets have sufficient stock (to_order_flag = False). This confirms the system is working correctly as a standing monitor — it will automatically trigger procurement recommendations when inventory falls below safety thresholds. The recommendation is to replace the uniform 40-unit reorder threshold with client- and asset-type-specific safety stock parameters derived from actual usage data.

8.2 User Value from the Project

- System Administrators gain complete operational visibility across all four modules from a single dashboard, with role-appropriate drill-down into ticket routing, SLA risk, hardware health, and inventory.
- Support Engineers receive AI-suggested resolutions and top 3 engineer assignments for each incoming ticket, reducing cognitive load and routing errors.
- Data Analysts can interrogate SLA compliance trends, breach probability distributions, model leaderboards, and feature importance through dedicated analytics views.
- Operations Managers can monitor SLA risk in real time, review AHP assignment suggestions before execution, and approve procurement recommendations with a 2-layer approval workflow.
- Executive Viewers receive synthesised strategic health reports: \$3.92M savings, 28.6% downtime reduction, 60.4% SLA compliance rate, quarterly milestone tracking — without requiring technical expertise.

8.3 Limitations and Future Work

- Maintenance History Gap: Asset-level maintenance data unavailable in the current dataset. Sponsor committed to tracking this data for future survival model iterations.
- Priority Classification Ceiling: Uniform P1–P4 distribution (~25% each) prevents meaningful priority prediction. Real data with genuine priority-feature correlations expected to improve accuracy to 65–75%.
- SLA Breach AUC on Synthetic Data: ROC-AUC ~0.50 is a synthetic data artifact. Real operations data expected to yield AUC 0.78–0.88 (confirmed in sponsor validation).

- **Live Telemetry Integration:** Current dashboard uses historical batch data. Real-time telemetry streaming from live infrastructure monitoring tools is required for production deployment.
- **BERT/Transformer Upgrade:** As ticket description completeness improves, upgrading from TF-IDF to Sentence-Transformer embeddings would substantially improve retrieval semantic depth.
- **Engineer Availability Real-time Feed:** Current AHP uses historical SLA success rates. Live engineer status feed from ticketing system required for real-time assignment accuracy.

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